
From Pretrained Knowledge to Action: Auditing Language Models in a Deterministic Game

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Abstract

1 Pretrained language models can describe strategies without reliably converting
2 that knowledge into actions. Measuring this gap is difficult because “planning”
3 benchmarks often change the task, interface, and degree of agent control at the
4 same time as the putative horizon. We present SLAY-BENCH, a deterministic
5 Slay-the-Spire-inspired simulator with four separately scored decision operations,
6 two characters, and two state encodings. Across seven open-model configurations,
7 differences are operation-specific. In the cleanest matched comparison, Qwen3-
8 235B-A22B improves fixed-deck card choice over Qwen3-32B in all four character-
9 encoding cells and 19/20 seed pairs, but not uniformly in immediate sequencing,
10 combat execution, or hybrid Act-1 rollout. An adversarial instrument audit then
11 finds that a non-planning card-name dictionary solves all 120 rotated fixed-deck
12 cases, while a constant answer solves the original removal task. These falsifications
13 bound what the current scores mean: they profile action-relevant recognition and
14 execution, not a common planning horizon. We retire the cross-task scalar and
15 specify a same-state controlled-horizon protocol; its model results are PENDING
16 EXPERIMENT. The broader lesson is methodological: before attributing failures
17 to an agent, benchmark designers should first try to solve their own instrument
18 cheaply.

19 1 The measurement problem

20 Pretraining may encode game rules, card interactions, and strategic abstractions, yet an acting agent
21 must select a legal action under a concrete objective and carry its consequences forward. PTA
22 therefore needs measurements that distinguish *stored knowledge*, *action selection*, and *closed-loop*
23 *control*. Existing planning benchmarks offer exact symbolic evaluation [10], while game benchmarks
24 offer interactive breadth [3, 8]. But a line drawn across heterogeneous tasks can look like a “planning
25 horizon” even when task semantics, output contracts, baselines, and agent authority all changed.

26 We use a deterministic deck-building game to ask a narrower question: *where does pretrained*
27 *knowledge become useful for a specified decision operation?* Slay the Spire is not new as an
28 LLM testbed. MiniSTS studied language-driven play [1]; subsequent work examines card-rule
29 synergy [2] and original-game agents across broader suites [9]. Our contribution is an inspectable,
30 within-simulator *operation profile* and the adversarial audit that changed its interpretation.

31 Three findings survive that audit:

- 32 1. a seven-configuration profile exposes operation-specific, not uniform, differences;
- 33 2. a matched Qwen3 comparison isolates a large card-choice gain (4/4 cells, 19/20 seed pairs);
- 34 and
- 35 3. compute-free shortcut policies invalidate the former cross-task horizon and removal claims.

Table 1: The four operations are not levels on one horizon. “Ref.” is the scoring reference; scripted choices are outside model control. Removal-v1 is quarantined.

Operation	Model-controlled output	Score / ref.	Supported interpretation
Immediate sequence	ordered card indices for one state	damage / exhaustive best	immediate-damage sequence quality
Combat execution	card plays each turn	win; remaining HP / greedy bot	closed-loop combat execution
Fixed-deck choice	deck label; one of three cards	authored label and pick	taxonomy and card-category alignment
Hybrid Act-1 rollout	combat and reward choices	floors and progress / matched greedy	behavior under scripted route, rest, shop, and events

2 A deterministic operation profile

SLAY-BENCH implements a deterministic, Slay-inspired simulator with independent random streams, two characters (Ironclad and Silent), Acts 1–3 mechanics, and two semantically matched prompt encodings: structured JSON and raw English. It is not claimed to be a bit-exact clone of the commercial game. Table 1 states what the model actually controls and the construct each score can support.

The immediate oracle exhaustively enumerates legal sequences; current-code replay completes all 200 paper-seed states below a declared 20,000-node budget (maximum 137 nodes). Combat uses a deterministic greedy reference, not optimal play. Fixed-deck fixtures contain one authored label and one on-label card among three rotated offers. The rollout is explicitly hybrid: the LLM chooses combat and rewards, while traversal and several non-combat decisions are scripted. Illegal actions and parse failures score as failures; card accuracies for unparseable outputs are not silently imputed.

Design and uncertainty. We evaluate seven configurations from five model families [6, 7, 4, 12, 11] with greedy decoding. For every available character–encoding cell, turn, combat, and fixed-deck tasks use 20 samples at each of five base seeds {42, 1042, 2042, 3042, 4042}; their sample streams do not overlap. Standard deviations are across base seeds. Hybrid runs use 20 runs per seed for the balanced small-model cohort and a separately reported 5-run floor-estimate tier for Qwen3; missing cells remain missing. Format tests use exact sign-flip permutations over matched seed pairs. With only five pairs, the minimum two-sided per-cell p is 0.0625, so inference relies on pooled strata and paired sample tests, not isolated null results.

3 Results: competence is operation-specific

Figure 1 reports native units rather than averaging non-commensurate scores. The profile immediately rejects a one-number account. DeepSeek-R1-Distill-14B has strong immediate sequencing but weak combat and a -2.73 -floor hybrid result in its one measured rollout cell. Small instruct models nearly match the greedy combat reference while differing materially in sequencing and card choice. Hybrid progress is usually within roughly one floor of seed-matched greedy; survival is at the same floor (0–3% in the balanced cohort), so we call it *on par*, not a win.

3.1 A selective Qwen3 capacity gain

Qwen3-235B-A22B (22B active parameters, FP8 MoE) and dense Qwen3-32B share a model family but not architecture, so this is a matched capacity comparison rather than a scaling law. Card-choice gains are $+0.256, +0.110, +0.280, +0.210$ for Ironclad structured, Ironclad raw, Silent structured, and Silent raw, respectively. Nineteen of 20 matched seed pairs improve. Archetype recognition rises in 15 pairs, ties in three, and falls in two. In contrast, turn differences mix signs ($+0.004, +0.043, -0.021, -0.031$), combat HP differences mix signs, and hybrid floor differences are $-0.64, -0.20, -0.12, +0.56$. Immediate sequencing is ceiling-limited (91–97/100 samples per cell already match the exact oracle), but that does not manufacture the card-choice result. The defensible conclusion is selective improvement in choosing a category-aligned card.

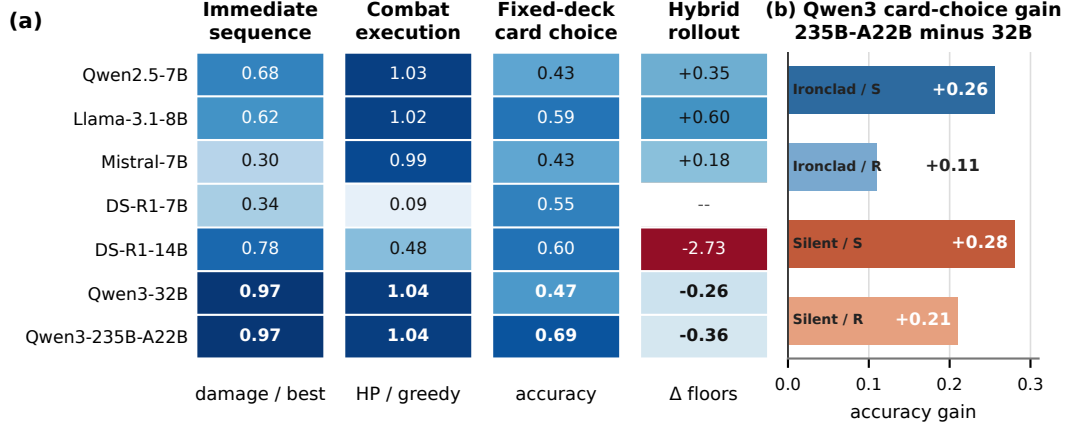


Figure 1: **Operation profiles, not a horizon curve.** (a) Each column has its own unit and scale; entries average the four character-encoding cells when available. Hybrid values are floors relative to a seed-matched greedy policy. DS-R1-14B hybrid is one cell; DS-R1-7B is unmeasured. (b) Qwen3-235B-A22B minus Qwen3-32B fixed-deck card-choice accuracy. S/R denote structured/raw. The MoE-versus-dense comparison is not a pure scaling law.

Table 2: Compute-free falsification tests over all fixed fixtures and rotations.

Target	Degenerate policy	Result	Consequence
Removal-v1	always answer “Strike”	40/40 fixtures	metric quarantined
Deck label + pick	card-name dictionary; choose sole on-label offer	60/60 per character	bounds score as recognition
Cross-task scalar	reweight incomparable task units	ranking is arbitrary	scalar and horizon curve retired

3.2 Encoding changes execution, not a general capability

Across 14 model-character strata, structured prompts improve combat win rate by 0.0479 (7 positive, 7 ties; pooled permutation $p = 10^{-4}$) and HP ratio by 0.0593 (12 positive, 2 negative; $p = 10^{-4}$). For immediate damage, however, the pooled magnitude favors raw by 0.0644 while directions split 6 versus 8. Fixed-deck effects also split by model. Thus representation format is part of the control interface, but its sign is not a domain-general property of “planning.”

4 Instrument falsification changes the claim

Shortcut learning is a measurement failure when a benchmark rewards a feature that does not require the intended capability [5]. We therefore ask the cheapest adversarial question: *what would a non-planning policy score?*

Every original removal fixture has the same expert answer, Strike. More subtly, each deck contains a unique signature card and each offer contains exactly one card assigned to that label. A deterministic dictionary therefore obtains perfect archetype and pick accuracy across 20 fixtures and three rotations per character (60/60 each; 120/120 combined), without simulating a transition. This does *not* show that the evaluated LLMs used the shortcut. It shows that the instrument cannot require planning. We consequently exclude removal from every analysis and composite, rename synergy as fixed-deck recognition/card choice, and retire the cross-task “horizon collapse” plot.

The audit also sharpens what is actionable. Qwen3’s capacity gain is real under the current labels, but the action is supported by recognizable card categories rather than demonstrated forward search. That distinction is precisely the gap between pretrained knowledge and acting that a PTA evaluation should expose.

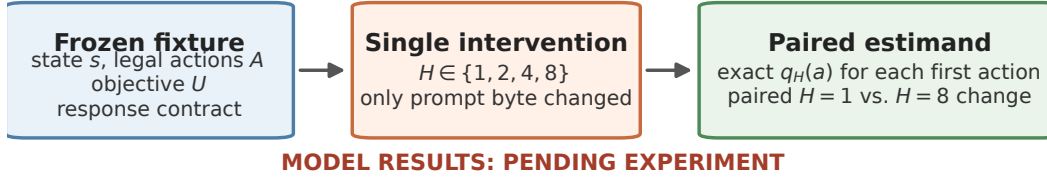


Figure 2: The proposed controlled decision-horizon test changes one variable within a fixture. Exact search scores every legal first action, avoiding cross-task normalization. Model results have not been run.

94 5 The controlled test we have not yet run

95 To identify decision horizon, we implement a versioned same-state protocol. From one fixture, hold
 96 observation, legal first actions, objective, response contract, simulator, and scoring fixed; vary only
 97 search horizon $H \in \{1, 2, 4, 8\}$. Exact search gives each legal first action’s H -step value. The
 98 primary estimand is the within-fixture change in first-action quality from $H = 1$ to $H = 8$, with the
 99 full slope reported. Fixtures are generated without model outputs and must pass three gates before
 100 inference: the optimal first action changes with H often enough to create treatment strength; exact
 101 search stays within an audited budget; and an H -mismatched greedy policy loses on the sensitive
 102 subset. A no-inference smoke test confirms prompt bytes differ only in H and the search does not
 103 mutate the fixture. Model results are PENDING EXPERIMENT; the protocol is a design, not evidence
 104 of longer planning.

105 **A reusable evaluation ladder.** The audit suggests four gates for claims about actionable represen-
 106 tations. First, test constant, lookup, and other compute-free policies. Second, report each operation in
 107 its native unit with the model’s authority made explicit. Third, intervene on the target construct while
 108 freezing the state and interface. Fourth, validate authored utilities against downstream outcomes or
 109 an external environment. SLAY-BENCH currently clears the first two gates, provides infrastructure
 110 for the third, and does not yet clear the fourth. This ladder makes the evidence status legible instead
 111 of hiding it in a scalar.

112 6 Limitations and conclusion

113 The simulator is deterministic but simplified and lacks conformance testing against the original game.
 114 Fixed-deck labels and utilities are authored rather than validated by downstream outcomes. Historical
 115 artifacts predate full provenance and run traces; DeepSeek-7B fixed-deck accuracies are conditioned
 116 on parseable outputs. The Qwen3 run tier has only 25 runs per cell, and the central Qwen3 comparison
 117 changes architecture as well as capacity. Seven open configurations, two characters, and one game do
 118 not establish general scaling or transfer. Finally, hybrid rollout does not measure a fully autonomous
 119 agent.

120 Within those limits, SLAY-BENCH supports a useful conclusion: pretrained knowledge does not
 121 become actionable uniformly. Capacity can improve a local card choice without uniformly improving
 122 execution or rollout, and a convincing “planning” profile can be produced by shortcuts in the
 123 instrument itself. Reporting operations in native units, testing degenerate policies, and reserving
 124 horizon claims for same-state interventions turn this negative audit into a constructive evaluation
 125 recipe.

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